

module4_lab

September 27, 2026

```
[1]: import pandas as pd
```

```
[2]: df=pd.read_csv("sales_clean.csv")
```

```
[3]: df.shape
```

```
[3]: (292, 6)
```

```
[4]: df.head()
```

```
[4]:
```

	order_id	date	product	price	qty	zip
0	1254	2026-04-06	mug	11.36	4	60614
1	1057	2026-05-30	charger	14.79	3	30303
2	1150	2026-05-06	mug	5.50	1	10001
3	1066	2026-05-30	notebook	37.53	1	98101
4	1114	2026-04-06	webcam	45.59	4	90405

```
[5]: print(df["price"].mean())  
print(df["price"].median())  
print(df["price"].std())
```

```
55.52315068493151
```

```
37.53
```

```
71.76708493504132
```

```
[6]: print(df["qty"].mean())  
print(df["qty"].median())  
print(df["qty"].std())
```

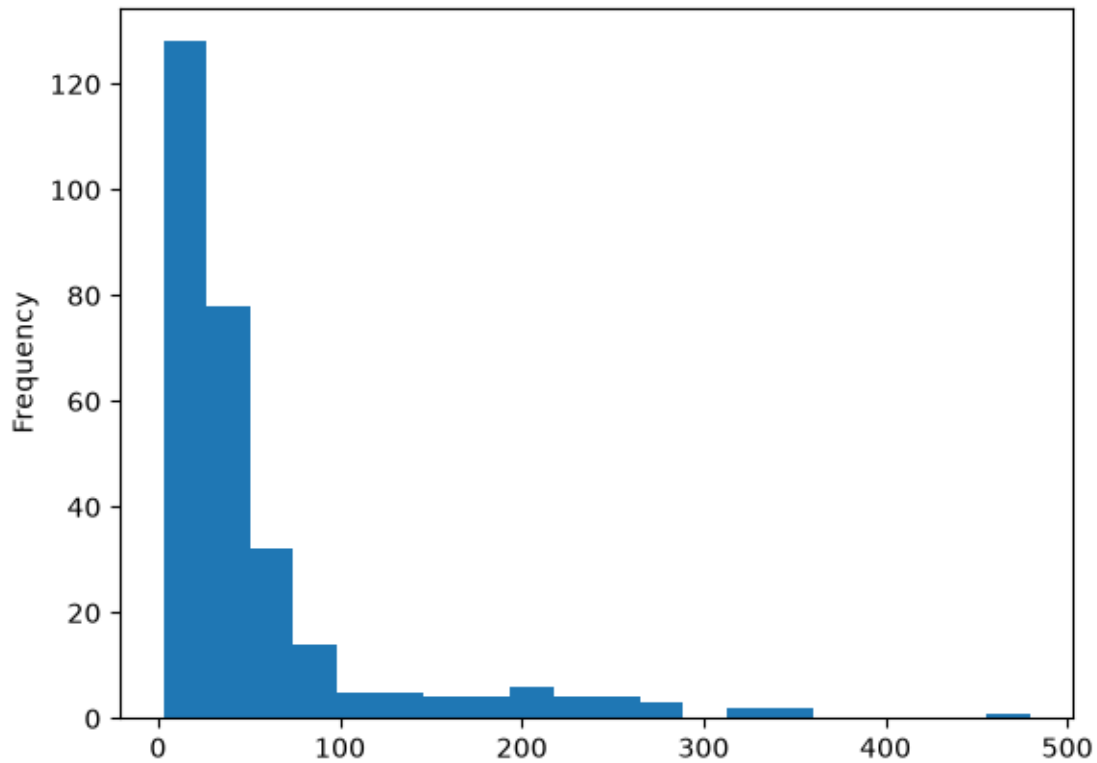
```
2.9863013698630136
```

```
3.0
```

```
1.4092785381929205
```

```
[7]: df["price"].plot(kind="hist", bins=20)
```

```
[7]: <Axes: ylabel='Frequency'>
```



```
[8]: df["revenue"]=df["price"]*df["qty"]
```

```
[9]: df.shape
```

```
[9]: (292, 7)
```

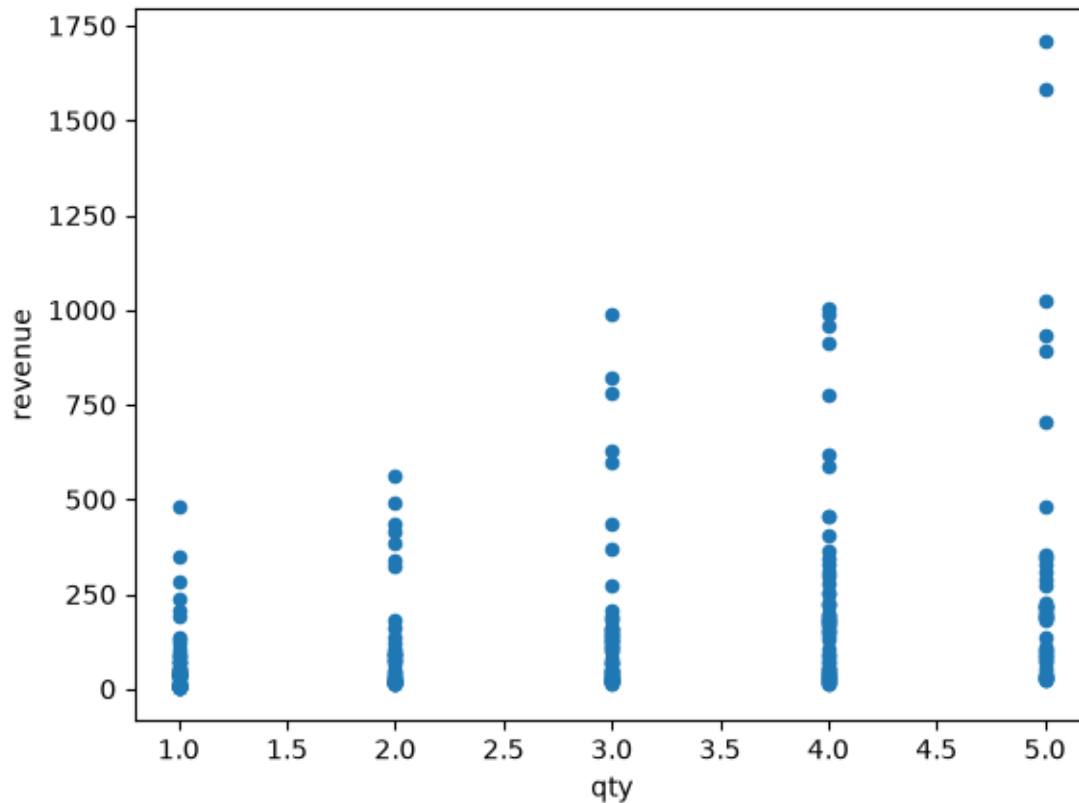
```
[10]: df.corr(numeric_only=True)
```

```
[10]:
```

	order_id	price	qty	zip	revenue
order_id	1.000000	-0.005070	0.031420	-0.014784	0.004181
price	-0.005070	1.000000	-0.041982	0.023812	0.821001
qty	0.031420	-0.041982	1.000000	0.086869	0.312793
zip	-0.014784	0.023812	0.086869	1.000000	0.047090
revenue	0.004181	0.821001	0.312793	0.047090	1.000000

```
[11]: df.plot.scatter(x="qty",y="revenue")
```

```
[11]: <Axes: xlabel='qty', ylabel='revenue'>
```



```
[12]: number_list=[2,4,4,6,9]
```

```
[13]: mean=pd.Series(number_list).mean()
      print(mean)
```

5.0

```
[14]: median=pd.Series(number_list).median()
      print(median)
```

4.0

```
[15]: #markdown cell
      # Using own corrected sales_clean.csv from module 3
      # I would report the median price since there is not much sensitivity to
      #   ↳ outliers like using the mean. Since expensive values on the skewed right
      #   ↳ histogram can skew the mean not giving a clear picture.
      #R value for quantity and revenue is positive meaning as more items are sold,
      #   ↳ we are making more money.
      #However, since our R value is not close to 1, there must be other factors at
      #   ↳ play beyond the amount of items sold
```

In addition, since we have a right skew histogram, there are potential
→ outliers that we would need to pay attention to